HydraWatch

Embedding-Based Wastewater Pathogen Surveillance for federated hospital networks

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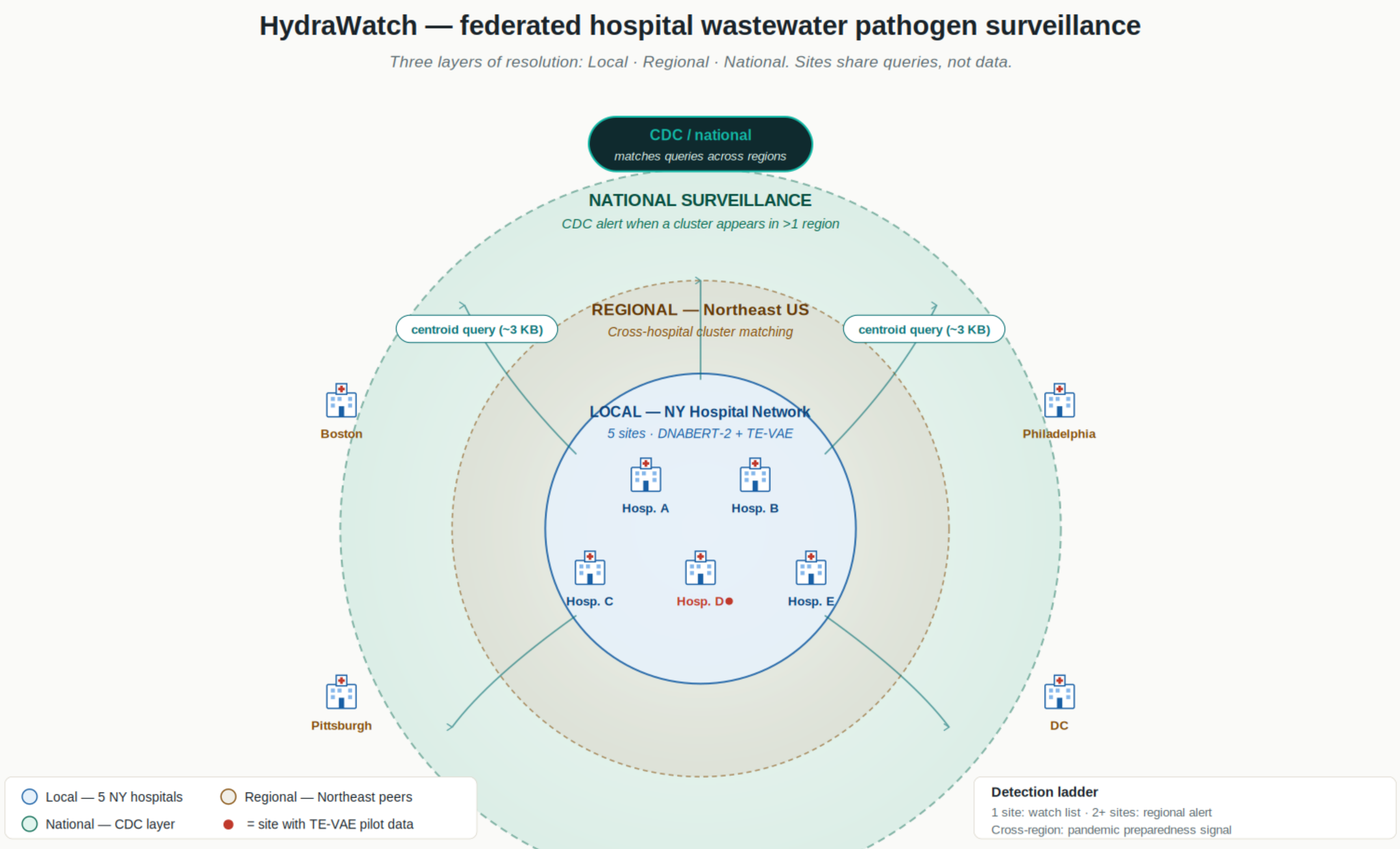
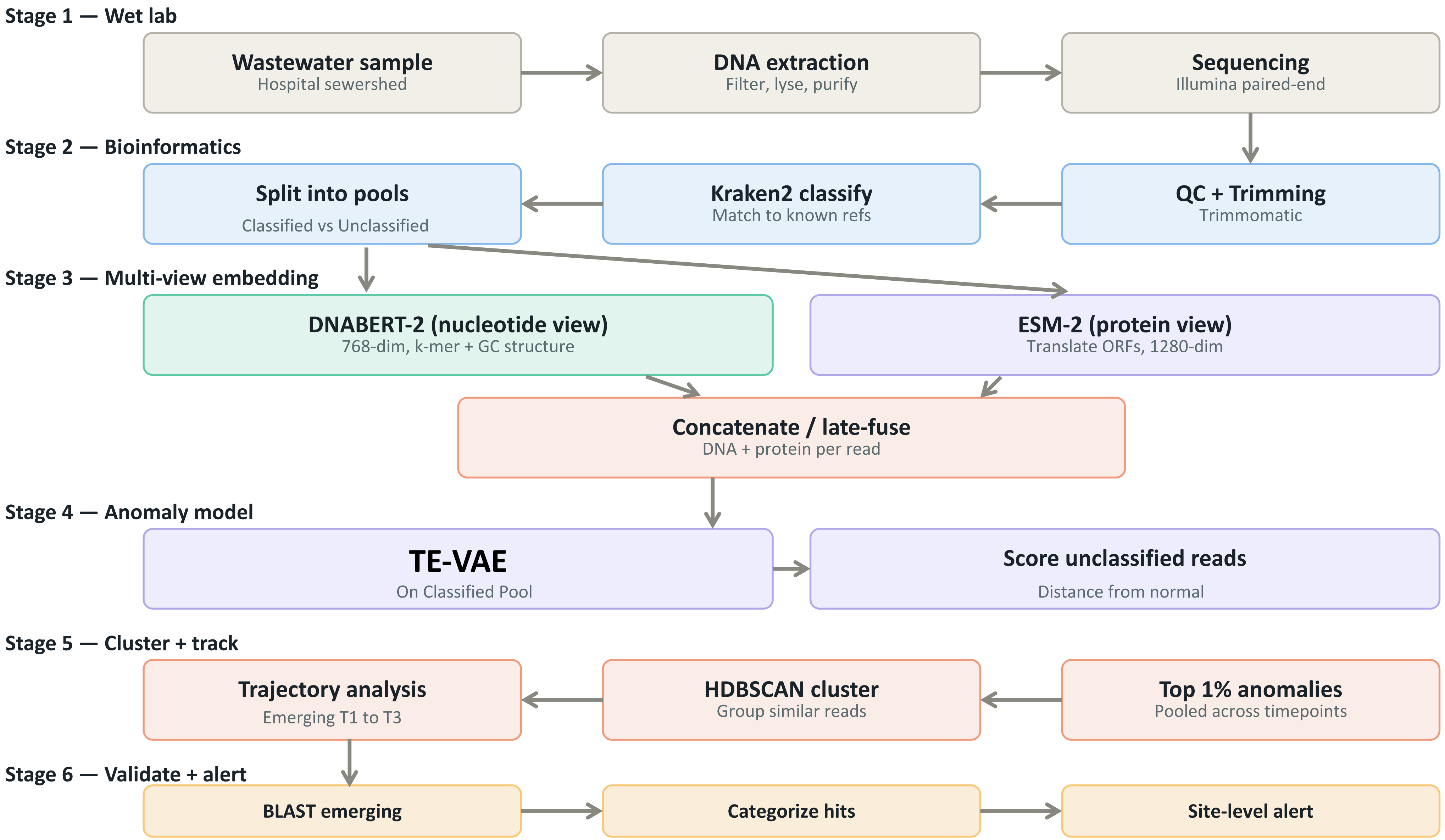
HydraWatch: Conceptual architecture



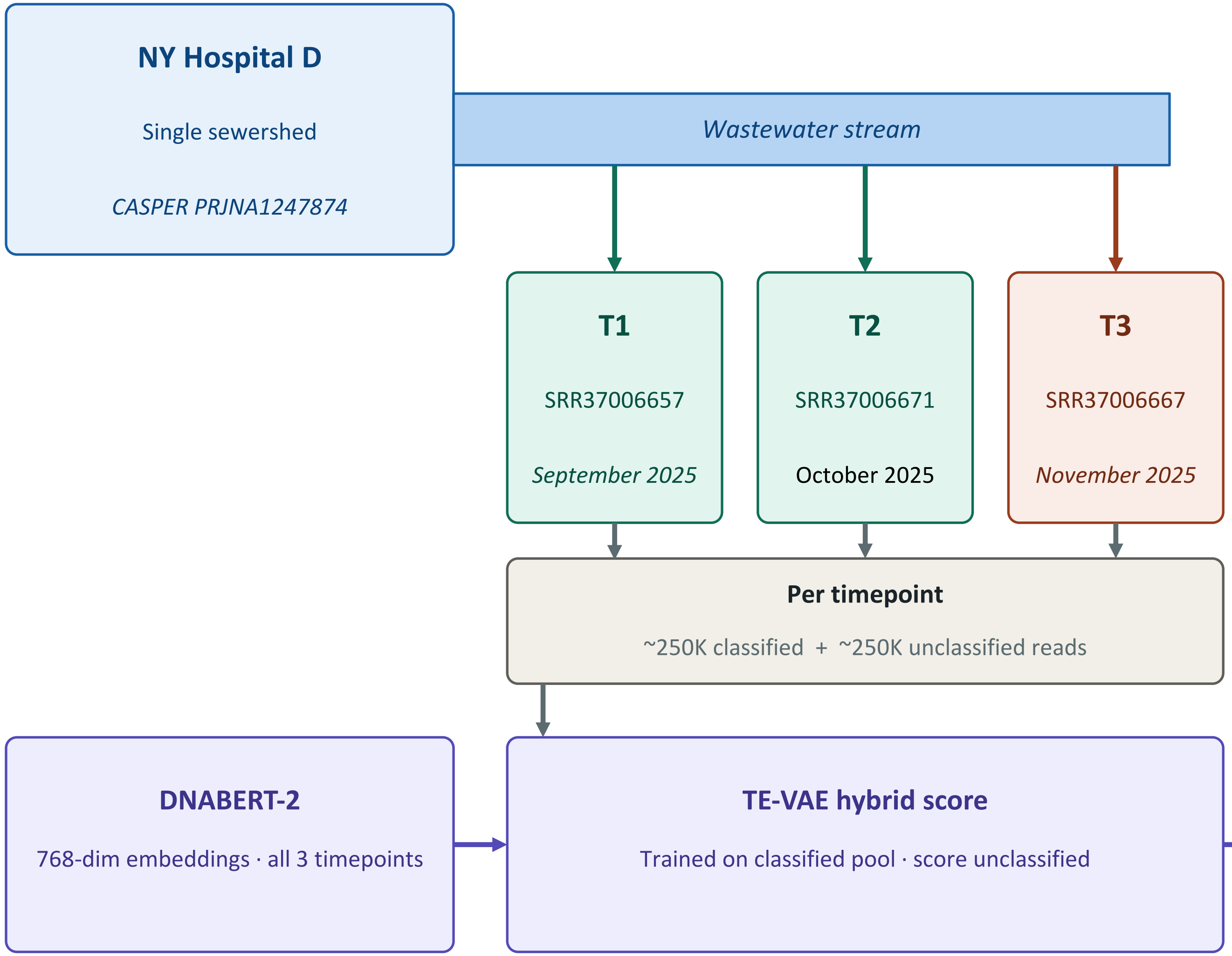
Privacy by design.

Raw reads and read-level embeddings stay on each hospital's servers. Sites exchange only cluster-centroid queries (~3 KB per emerging cluster), sidestepping the data-sharing agreements that typically slow multi-site surveillance.

HydraWatch: Data flow and processing stages



HydraWatch: Longitudinal Sampling design: NY Hospital D, three timepoints



Cluster trajectories across T1 → T3

Emerging vs transient · early-warning candidates

Why one site, three timepoints

Same sewershed = same baseline biology.
Anomaly scores comparable across time → can call emerging vs persistent vs transient signals.

What we found

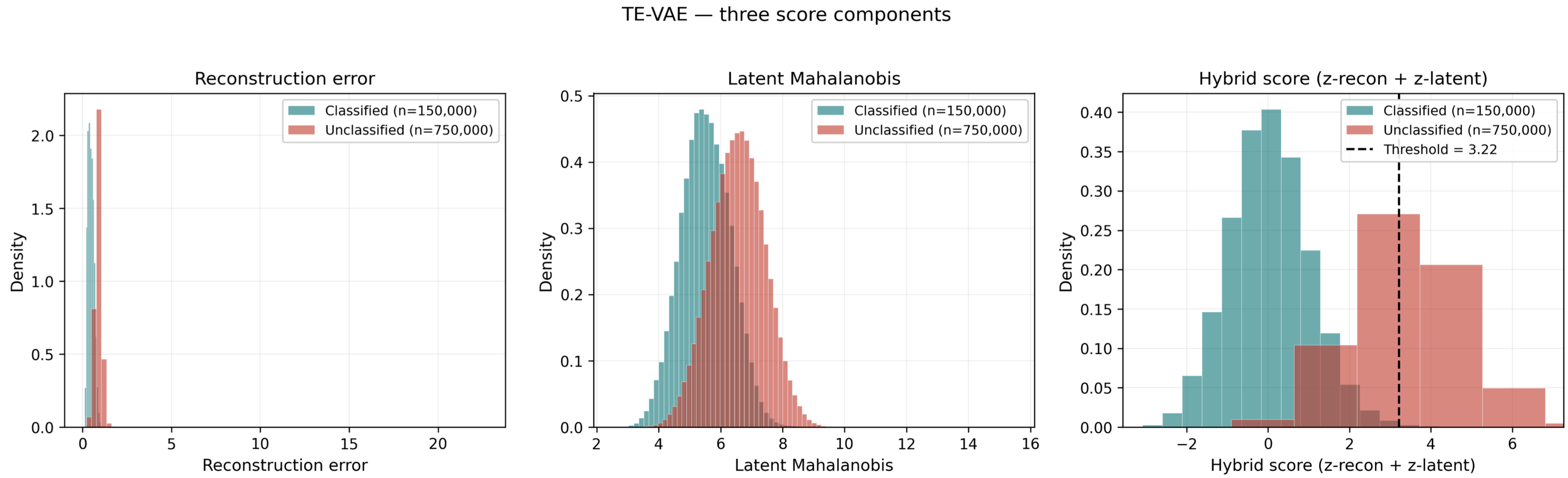
Emerging signal at T3
Cluster 6: 284 → 122 → 3,506 reads (×12.3 growth)

Trajectory diversity
Transient T2 and declining T1 clusters separated naturally

BLAST validation
Representative reads queued for follow-up

Same site, same baseline model, three timepoints → emerging clusters become an early-warning candidate signal.

Results: Multi-timepoint anomaly detection — DNABERT-2 + TE-VAE



Key finding

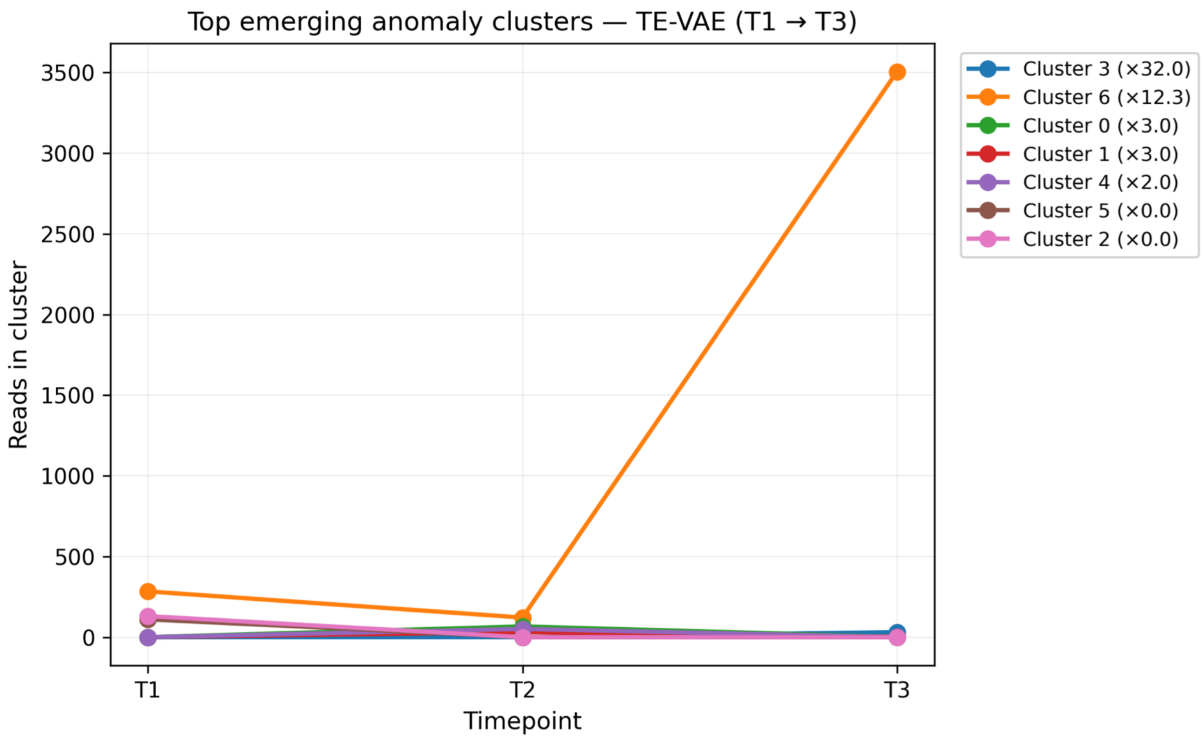
Two emergence patterns at T3:

Cluster 6: ×12.3 growth, 3,506 reads: the dominant signal.

Cluster 3: zero at T1/T2, 31 reads at T3: clean emergence at low mass.

Method: DNABERT-2 embeddings → TE-VAE hybrid score (reconstruction + latent Mahalanobis) on pooled classified pool → HDBSCAN on top 1% across all timepoints.

Results: Emerging Cluster across timepoints



Cluster	T1	T2	T3	Growth	Pattern
6	284	122	3506	x12.3	Emerging
3	0	0	31	x32	Emerging (low mass)
1	0	31	2	x3	Transient (T2 only)
5	111	0	0	X0.009	Declining (T1 only)

Key finding

Two emergence patterns at T3

Cluster 6: *×12.3 growth, 3,506 reads — the dominant signal.*

Cluster 3: *zero at T1/T2, 31 reads at T3 — clean emergence at low mass*

Trajectory analysis separates emerging, transient, and declining clusters; cluster 6 representative reads queued for BLAST validation.

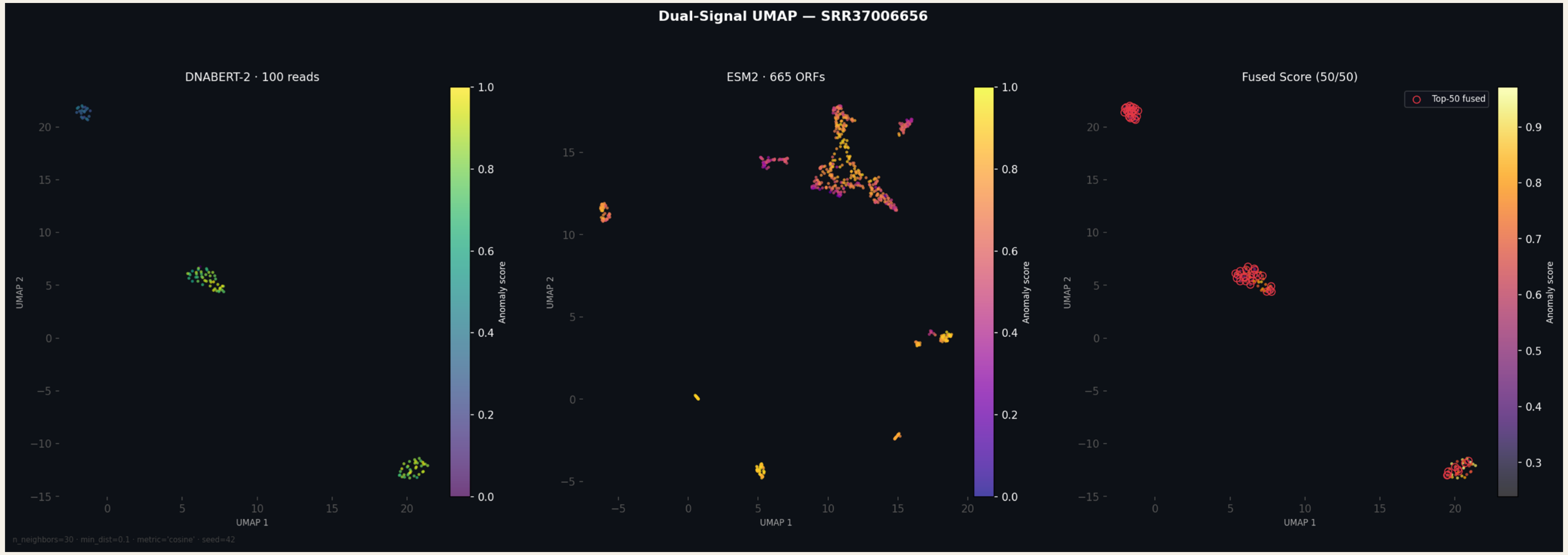
Results: Multi-view embedding: DNABERT-2 + ESM-2 (one timepoint)

Why two views

DNABERT-2 sees nucleotide structure: k-mer composition, GC content, codon-like patterns.

ESM-2 sees protein structure: it translates the read's open reading frames and embeds the protein sequence directly.

Together they catch anomalies either view alone would miss — protein-coding novel pathogens with divergent DNA but conserved protein folds become visible.



Nucleotide view DNABERT-2 — 768-dim per read

Protein view ESM-2 — 1280-dim per ORF

Late-fusion Concatenate → joint anomaly score

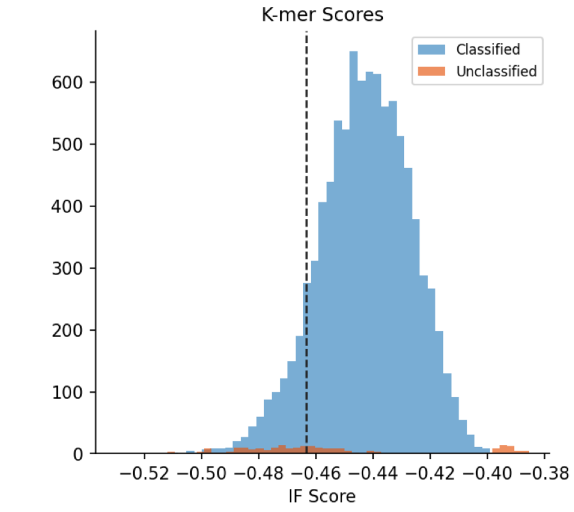
Embedding space sees structure references can't

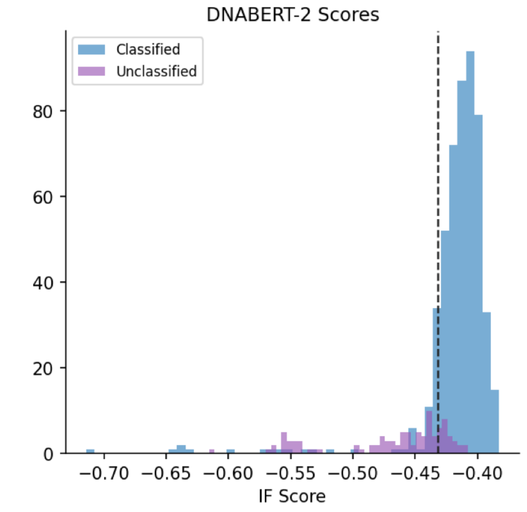
Top-50 fused-signal reads (red rings) cluster coherently across DNA, protein, and fused embedding spaces. SRR37006656 (CASPER, 5 Nov 2025).

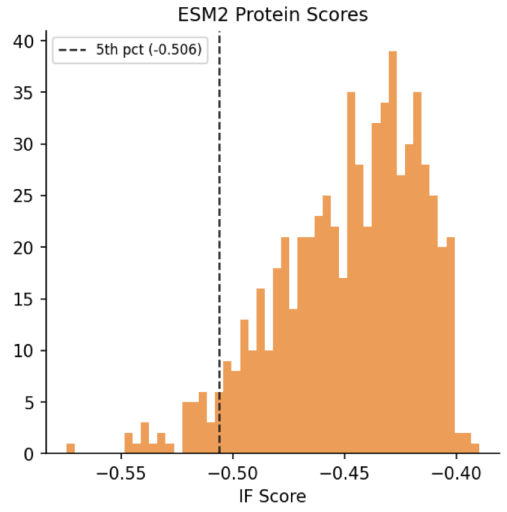
Multi-view embeddings extend the single-view analysis — current longitudinal implementation uses DNABERT-2 only; ESM-2 with timepoints fusion could be a next step.

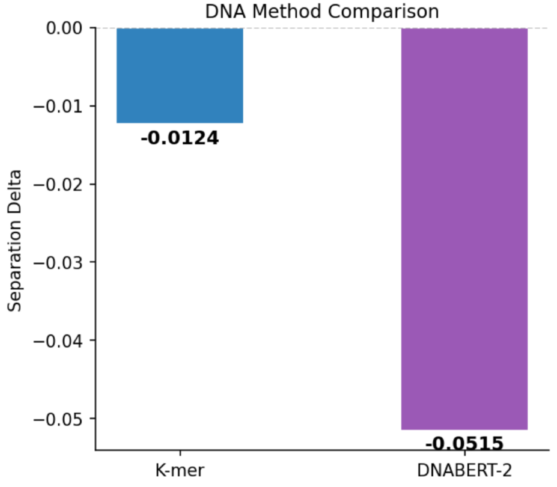
Results: Multi-view embedding: DNABERT-2 + ESM-2 (one timepoint)

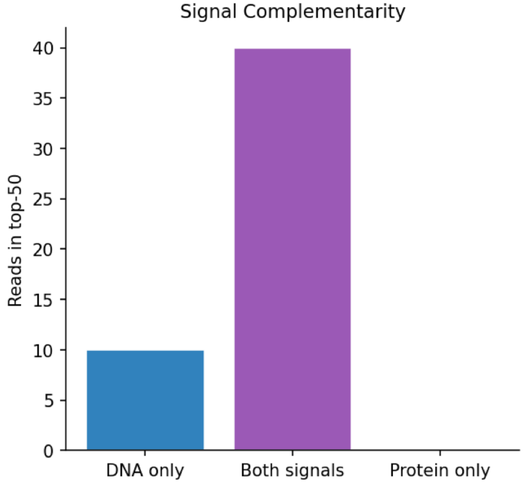
Unified Pipeline Summary — SRR37006656 | Unclassified: 2.4% | K-mer delta=-0.0124 | DNABERT-2 delta=-0.0515 | ESM2 tail=5.1% | Verdict: ▯ MONITOR











BLAST Results

Mode	%id	E-val	Triage	Hit
BLASTn	98.0%	1e-39	▯ ROUTINE	Brevundimona
BLASTn	73.3%	7e-10	▯ MEDIUM	Azospirillum sp
BLASTn	76.7%	3e-09	▯ MEDIUM	PREDICTED: Ga
BLASTn	92.0%	6e-49	▯ ROUTINE	Mutant Gallid a
BLASTn	92.7%	5e-50	▯ ROUTINE	Mutant Gallid a
BLASTp	0.0%	1e+00	▯ MEDIUM	No hit
BLASTp	0.0%	1e+00	▯ MEDIUM	No hit
BLASTp	77.3%	7e-09	▯ MEDIUM	Uncharacterise
BLASTp	0.0%	1e+00	▯ MEDIUM	No hit
BLASTp	96.0%	8e-17	▯ ROUTINE	autotransporte

Key finding

40 of top 50 reads flagged by both DNA and protein signals. BLAST recovers known organisms at high identity.

Validation: do embeddings recover what reference pipelines find?

The validation question

If our pipeline truly captures wastewater biology, it should recover known pathogens — even though they came from the ***unclassified*** pool that Kraken2 missed.

What we did

1. Took top emerging clusters (6 and 3) at T3
2. Extracted 5 representative reads per cluster (top by hybrid anomaly score)
3. Submitted to NCBI blastn (somewhat-similar mode)
4. Hits to be cross-referenced against the CASPER pathogen list

Result

Status: BLAST submission queued; results in follow-up.

The trajectory pattern itself : emergence concentrated at T3 with ×12.3 growth: is the signal HydraWatch is designed to surface.

Sequence-level anchoring is the next validation layer.

Three validation buckets

A — **CASPER pathogen recovered**

Validates: pipeline finds known biology in unclassified reads

B — **Known organism, not CASPER**

Real biology Kraken2 missed (e.g., environmental protists)

C — **No confident hit**

Candidate dark matter — coherent embedding clusters absent from NCBI

Limitation: 'no NCBI hit' reflects database coverage, not confirmed novelty: wastewater contains many organisms with no sequenced close relatives. tblastx + assembly would distinguish divergence from true novelty.

Conclusion: How embeddings complement SecureBio's pipeline

SecureBio's pipeline	HydraWatch adds	Why it matters
Kraken2 + reference DB	**Foundation-model embeddings**	*Sees biology even when no reference exists*
Tracks known pathogens over time	**Detects emerging clusters in unclassified reads**	*Surfaces threats before they're characterised*
Reference-based growth detection	**Reference-free anomaly scoring**	*No database to update; works on day one of a new outbreak*
Site-level analysis only	**Federated centroid queries across sites**	*Cross-site signal without sharing raw patient-derived reads*

The novelty

Reference-free

Anomaly score doesn't depend on a database — surfaces what isn't known yet

Privacy-preserving

Only cluster centroids (~3 KB) leave the site; raw reads and read-level embeddings stay on hospital premises.

Multi-site capable

Same embedding space works across hospitals → nationwide outbreak signal

Trajectory-aware

Distinguishes emerging, transient, and declining clusters : the early-warning signal is anything new or accelerating.

HydraWatch is complementary to — not a replacement for — reference-based surveillance. The two layers cover different failure modes.